Function and Struct

Fundamentals of Programming – CSC10012

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Contents

- Function.
- Multiple-file project.
- struct.

Function

- Problem with repeated code:
  - Consider a program:

    Enter 3 positive integers a, b, c >= 0.

    Compute and print S = a! + b! + c!.

    ➔ Identify repeated code.
  - Disadvantages of repeated code:

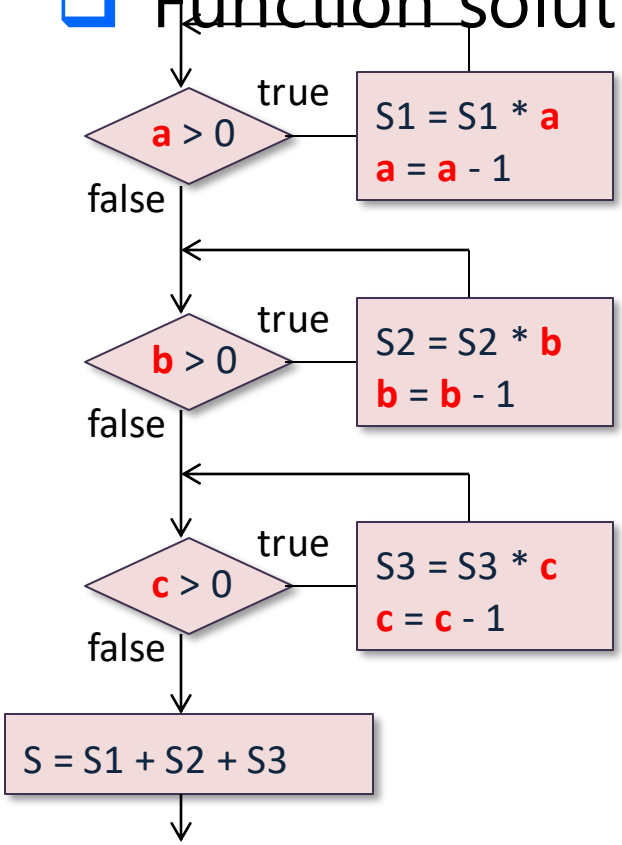
    Time and cost.

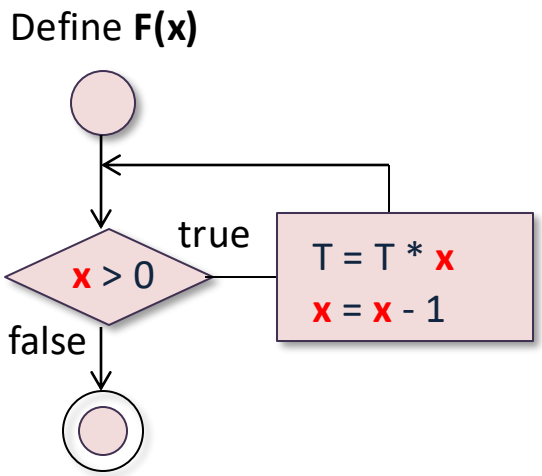
    Changes → fix multiple places.

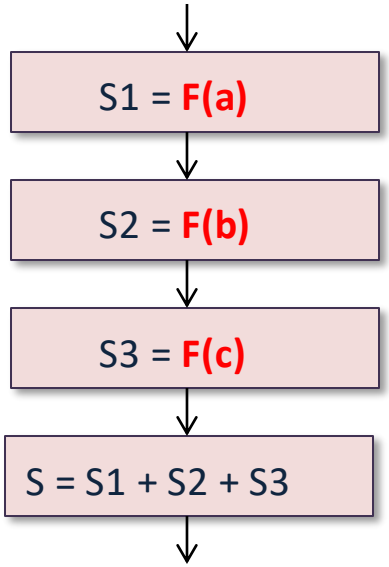
    ➔ Write once, reuse everywhere.

Function

- Function solution:

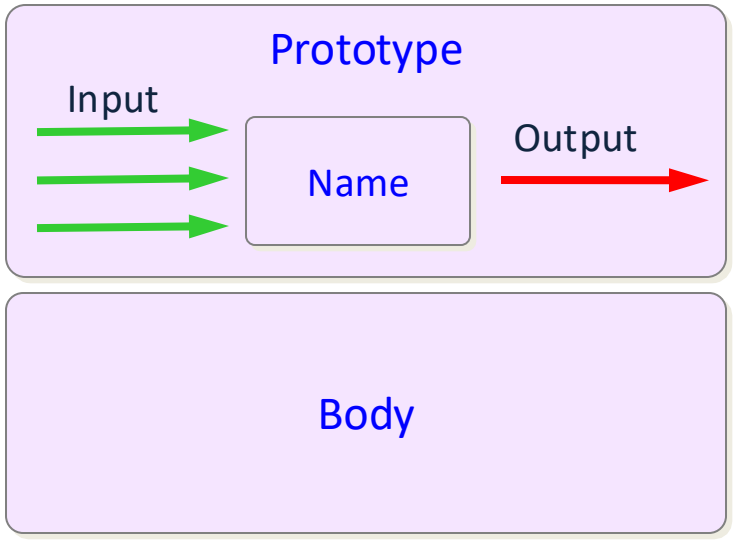






Function

- C/C++ function:
  - A named block of statements.
  - Can be called:
    - From anywhere in program.
    - With different arguments.
  - Function structure:
    - Prototype: declaration.
      - Function name.
      - Arguments.
      - Return type.
      - ➔ Identification.
    - Body: implementation.



Function

- C/C++ function:
  - Declaration (prototype):

    <*Return type*> **<Function name>(** <*Arguments*> **)**;

    <Return type>: int, float, char, ..., **void** (không trả về).

    float **calcGPA(** float literature, float math **)**;

    void **printResult**( );
  - Implementation (body):

    <*Return type*> **<Function name>(** <*Arguments*> **)**

    {

    [Statements]

    [**return** <*value*>;]

    }
  - Calling:

    <Function name>(<*Arguments*> **)**;

    float gpa = **calcGPA**(7.0, 8.5);

Function

- C/C++ function:

```
// Function declaration.
long long factorial( int n );

int main()
{
	// Declare and input a, b, c.

	// Function calls.
	S1 = factorial( a );
	S2 = factorial( b );
	S3 = factorial( c );

	S = S1 + S2 + S3;
}
```

```
// Function implementation.
long long factorial( int n )
{
	long long s = 1;
	for ( ; n > 0; n--)
		s = s * n;
	return s;
}
```

Function

- Passing arguments:
  - Pass-by-value:

    Argument values are passed to function.

    Function receives only the COPY.

    Real arguments are UNCHANGED.

    Arguments are: variables, constant, expressions.

```
float calcGPA( float lit, float math )
{
    lit = lit * 2;
    math = math * 3;
    return (lit + math) / 5;
}
```

```
int main()
{
    float a, b, gpa;

    gpa = calcGPA( a, b );
    gpa = calcGPA( 6, 8.5 );
    gpa = calcGPA( a + 1, b );
    // a, b are UNCHANGED.
}
```

Function

- Passing arguments:
  - Pass-by-reference (C++):

    Real arguments are passed to function.

    Function receives the original ones.

    Real arguments can be CHANGED.

    Arguments are variables only.

    Syntax: **&**<argument>.

```
float calcGPA( float &lit, float math )
{
    lit = lit * 2;
    math = math * 3;
    return (lit + math) / 5;
}
```

```
int main()
{
    float a, b, gpa;

    gpa = calcGPA( a, b);
    // a is CHANGED.
    gpa = calcGPA( 6, 8.5 );      //wrong
    gpa = calcGPA(a + 1, b);      //wrong
}
```

Function

- Passing arguments:
  - Notes:

    Use pass-by-reference to return values.

    ➔ Function with multiple return values.

```
void input( float &lit, float &math )
{
        printf("Enter literature = ");
        scanf("%f", &lit);
        printf("Enter math = ");
        scanf("%f", &math);
}
void calcGPA( float lit, float math, float &gpa )
{
        lit = lit * 2;
        math = math * 3;
        gpa = (lit + math) / 5;
}
```

```
int main()
{
        float  a, b;
        float  gpa;

        // a, b are UPDATED.
        input(a, b);

        // gpa are UPDATED.
        calcGPA(a, b, gpa);
}
```

Function

- Scope:
  - Existing area of variables and functions.

    Global scope: across program.

    Local scope: only in declaration block.
  - Function has global scope.
  - Variable:

    Global variable: declared outside functions (includes main()).

    ➔ Can be used across program.

    Local variable: declared inside a block.

    ➔ Can be used only in the block.

Function

```
float  S;                          // Global declarations.
int compute();

int main()
{
        int  a = S + compute();        // Local variable in main.
        while (a > 0)
        {
                int  b = S + compute();   // Local variable in loop.
        }
}

int compute()
{
        int  y = S * 2;                    // Local variable in function.
}
```

Contents

- Function.
- **Multiple-file project.**
- struct.

Multiple-file project

- How do we organize a book?
  - Cannot write in one paper!!
    - ➔ Split into chapters.
    - ➔ Summary at first.
    - ➔ Chapter contents follow.

Multiple-file project

- Organize C/C++ project:
  - Like a book:

    Chapters ~ source code files.

    Summary ~ main() function.

    ➔ How to connect multiple source code files?

```
// File main.cpp
int main()
{
        input();
        compute1();
        compute2();
        output();
}
```

```
// File io.cpp
void input()
{
}

void output()
{
}
```

```
// File compute.cpp
int compute1()
{
}

int compute2()
{
}
```

Multiple-file project

- Header file:
  - Connect source files across project.
  - Make code on a file "see" code on another file.
  - File extension **.h**.
  - Usage:

    Create header file **.h** for source file **.cpp**.

    File **.h** contains only **declaration** (global variables/functions).

    File **.cpp** contains **implementation** of functions.

    To let A.cpp "see" code in B.cpp

    ➔ A.cpp **#include "**B.h**"**

Multiple-file project

```
// File main.cp
#include "io.h"
#include "compute.h"

int main()
{
        input();
        compute1();
        compute2();
        output();
}
```

```
// File io.h
// Function declaration
void input();
void output();
```

```
// File compute.h
// Function declaration
int compute1();
int compute2();
```

```
// File io.cpp
#include "io.h"
void input()
{
}
void output()
{
}
```

```
// File compute.cpp
#include "compute.h"
int compute1()
{
}
int compute2()
{
}
```

Multiple-file project

- Divide-conquer a project:
  - How to eat a cow?

    ➔ Split into small parts.

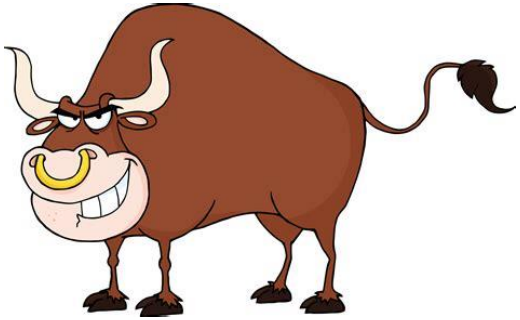
    ➔ Eat each parts.
  - How small is small?

    ➔ Can be chewed.
  - Organize a project:

    Split into functions and files.

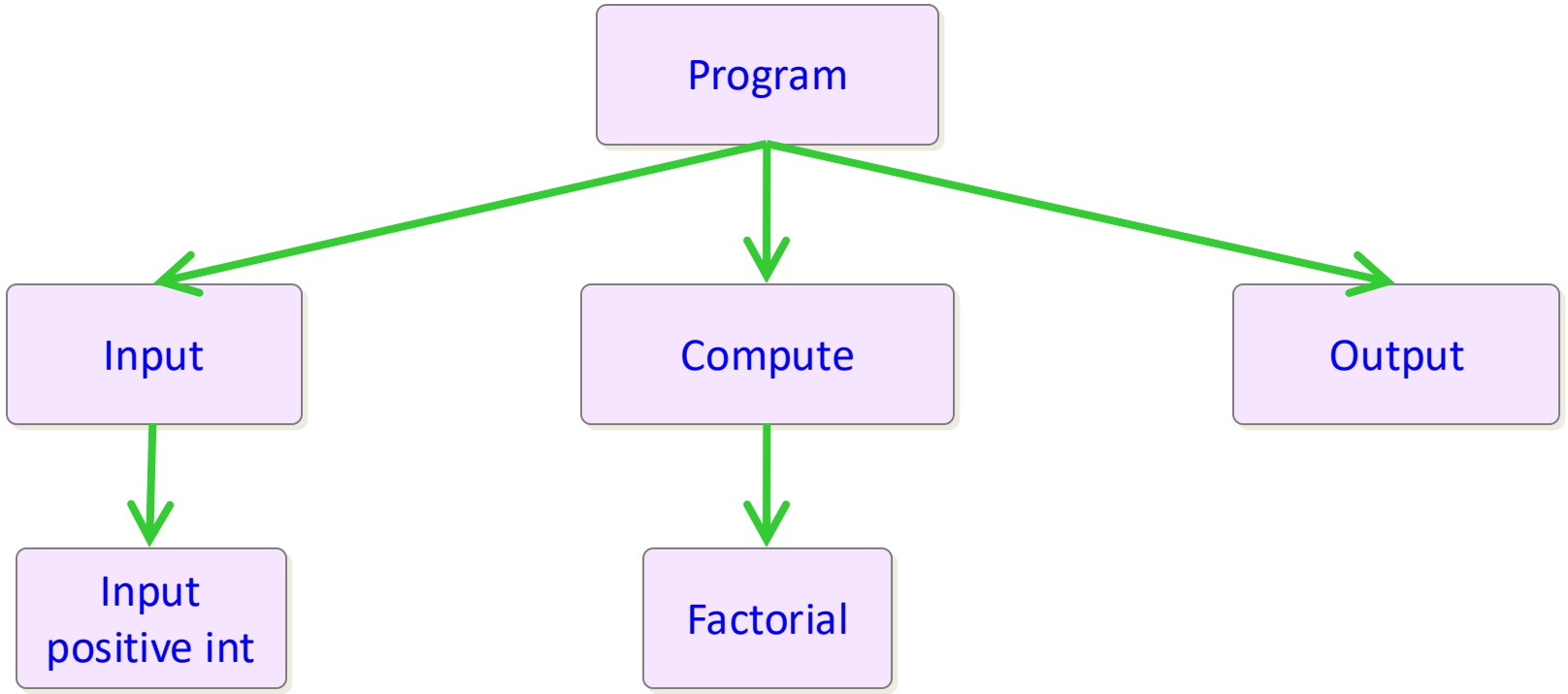
    Implement each function.

    Should be < 30 statements.



Multiple-file project

- Program breakdown tree:
  - Enter 3 positive integers a, b, c >= 0.
  - Compute and print S = a! + (b + 1)! + (c + 2)!.



Multiple-file project

```
// File main.cpp
#include "process1.h"

int main()
{
        int     a, b, c;
        long S;

        input(a, b, c);
        S = compute(a, b, c);
        output(S);
}
```

```
// File process1.h
void input(int &a, int &b,int &c);
long compute(int a, int b, int c);
void output(long result);
```

```
// File process1.cpp
#include "process1.h"
#include "process2.h"
void input(int &a, int &b,int &c)
{
        input_num(a);
        input_num(b);
        input_num(c);
}
long compute(int a, int b, int c)
{
        return factorial(a) +
factorial(b+1) + factorial(c+ 2);
}
void output(long result)
{
        printf("S = %ld", result);
}
```

```
// File process2.h
void input_num(int x);
long factorial(int n);
```

```
// File process2.cpp
#include "process2.h"
#include <stdio.h>
void input_num(int &x)
{
   do {
        printf("Positive integer = ");
        scanf("%d", &x);
   } while (x < 0);
}

long factorial(int n)
{
        long S = 1;
        for ( ; n > 0; n--)
                S = S * n;
        return S;
}
```

Contents

- Function.
- Multiple-file project.
- **struct.**

struct

- Organize program data:
  - Student information:

    Student id.

    Student name.

    Literature, math.
  - Write program:

    Enter information of a student.

    Print student information with computed gpa.

    ➔ What is the inconvenience?

struct

- C/C++ struct:
  - Combine data into one place.
  - Compound data-type.
  - Declaration:

```
// Syntax 1:
struct <struct name>
{
  <struct member 1>;
  <struct member 2>;
  ...
};
```

```
// Syntax 2:
typedef struct
{
        <struct member 1>;
        <struct member 2>;
        ...
} <struct name>;
```

struct

- C/C++ struct:
  - struct variable:

    struct <struct name> <variable>;

    // With typedef struct or C++
    <struct name> <variable>;

```
struct Student
{
   char  id[9];
   char  name[50];
   float  literature;
   float  math;
};

int main()
{
   struct Student  s1;
   Student  s2;  // C++
}
```

struct

- C/C++ struct:
  - Initialization:

```
struct <struct name> <variable> =
{   // In declaration order.
    <member 1 value>,
    <member 2 value>,
  ...
};
struct <struct name> <variable> =
{   // C99 standard.
    .<member name> = <value>,
    ...
};
```

  - Access struct member:

<variable> **. <member name>**.

```
int main()
{
    struct Student s =
    {
        "24127001",
        "Nguyen Van A",
        7.5,
        8.0
    };

    s.literature = 5.5;
}
```

struct

- C/C++ struct:
  - Passing arguments:

    Pass-by-value:

    ➔ Pass a copy.

    ➔ Unchanged.

    Pass-by-reference (C++).

    ➔ Pass the real one.

    ➔ Can be changed.

```
void  add1( struct Student s )
{
        s.literature++;
        s.math++;
}
void  add2( Student &s )
{
        s.literature++;
        s.math++;
}
int main()
{
        struct Student  s1, s2;
        add1( s1 );
        add2( s2 );
        // s1 unchanged.
        // s2 changed.
}
```

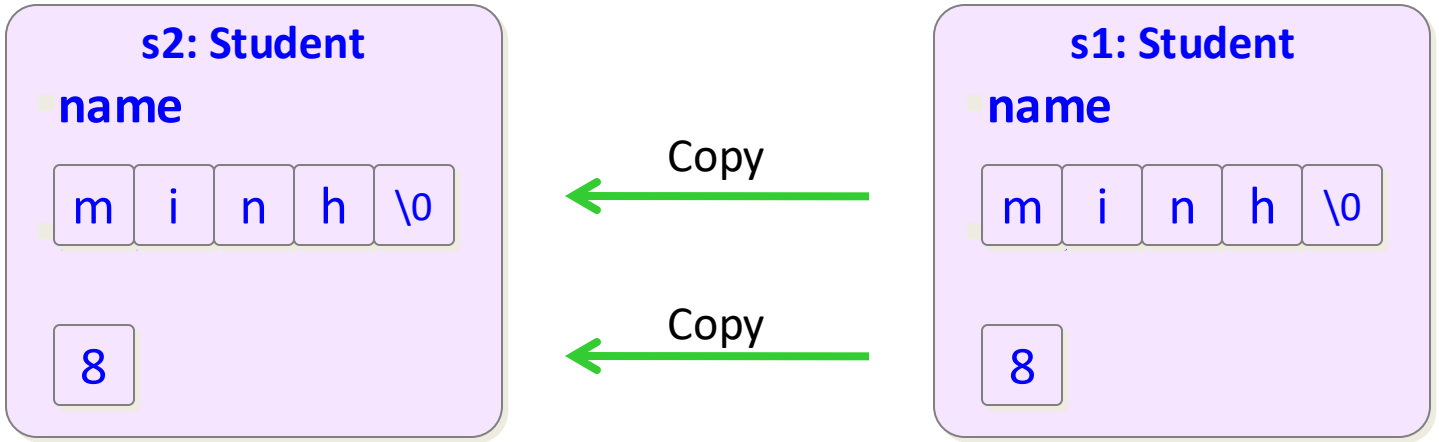
struct

- C/C++ struct:
  - struct assignment:

    All struct members are copied.

    Array members are copied too!

```
struct Student  s1 = { .name = "minh", .literature = 8.0 };
struct Student  s2 = s1;
```



Summary

- Function:
  - A named block of statement can be called anywhere.
  - Function structure:

    Prototype: name, arguments, return type.

    Body: implementation.
  - Passing arguments:

    pass-by-value.

    pass-by-reference (C++).
- Multiple-file project:
  - Program ~ book:

    main() ~ summary.

    source files ~ chapters.

Summary

- Multiple-file project:
  - Header file .h: connect code across source files.
  - Program breakdown tree:
    - Split program into files and functions.
    - Based on levels of abstraction and reusability.
- struct:
  - Compound data-type.
  - Combine data into one place.
  - Assignment: all members are copied.

Practice

- Practice 4.1:

  Write C/C++ program to find prime numbers:
  (organize in functions and multiple-file project)

  - Enter a positive integer N (re-enter if invalid).
  - Print all prime numbers <= N.

  Input format:

  Enter a positive integer = 11

  Output format:

  #1 = 2
  #2 = 3
  #3 = 5
  #4 = 7
  #5 = 11
  There are 5 prime numbers.

Practice

- Practice 4.2:

Write C/C++ program to simulate a calculator as follow:

- Enter two integers.
- Enter an operator (+, -, *, /, %).
- Perform the operator on two integers and print result.

Notes: flush the standard input stream after each input.

- C: fgets, or while getchar, C++: cin.getline, or cin.ignore.

Input format:

Enter two integers = 7 5

*Enter an opertor (+, -, *, /, %) = +*

Output format (no error):

Result = 12

Output format (divided-by-zero error):

Error: divided by zero.

Practice

- Practice 4.3:

  Write C/C++ program to classify a triangle:

  (organize in functions and multiple-file project)

  - Enter 3 positive real numbers a, b, c (re-enter if invalid).
  - Check if a, b, c can forms a triangle.
  - If yes, print the triangle type.
    (normal, right, isosceles, right-isosceles, equilateral).

  Input format:

  Enter 3 positive real numbers = 3 4 5

  Output format (can form a triangle):

  Can form a triangle.

  Right triangle.

  Output format (cannot form a triangle):

  Cannot form a triangle!

Practice

- Practice 4.4:

  Write C/C++ program to simulate a menu as follow:
  (organize in functions and multiple-file project):

  - Print the menu:

    1. Practice 4.1.

    2. Practice 4.2.

    3. Practice 4.3.

    4. Exit.

    Selection (1-4):

  - Enter an integer for your selection.
  - Selection 1-3:
    + Execute the selected practice.
    + Go back to menu.
  - Selection 4: exit program.

Practice

- Practice 4.5:

  Write C/C++ program to operate on fractions:

  (organize in functions and multiple-file project):

  - Declare struct to represent a fraction.
  - Enter 2 fractions.
  - Perform the following operations on 2 fractions and print result: add, multiply, inverse, reduce.

Practice

- Practice 4.6:

  Write C/C++ program to operate on students:

  (organize in functions and multiple-file project):

  - Declare struct to represent a student (stated in the lesson).
  - Enter a students.
  - Print GPA and their rank:
    + Excellent: GPA >= 8.5.
    + Good: GPA >= 7.0.
    + Fair: GPA >= 5.0.
    + Failed: GPA < 5.0.

Practice

- Practice 4.7:

  Write C/C++ program to operate on triangle:

  (organize in functions and multiple-file project):

  - Declare structs to represent point (x, y) and triangle (3 points).
  - Enter information of a triangle.
  - Compute and print triangle perimeter.
  - Find and print triangle centroid.

Practice

- Practice 4.8:

  Write C/C++ program to operate on date:

  (organize in functions and multiple-file project):

  - Declare struct to represent date (day, month, year).
  - Enter two date d1 and d2.
  - Check if d1 is latest than d2 and print result.
  - Print tomorrow date of d1.
  - Print yesterday date of d2.

Practice

- Practice 4.9 (*):

Simple project with multiple folders	Simple project with external libraries	Complex project project 1 uses project 2
project/ bin/ src/ subfolder/	project/ bin/ lib/ libA/ libB/ src/ subfolder/	project/ lib/ libA/ libB/ subproject1/ bin/ src/ subfolder1/ subproject2/ bin/ src/ subfolder2/

Q
&
A